Jonathan Cantu

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Data Science Analyst and Software Developer

SUMMARY

Accomplished Data Science Analyst and Software Developer with hands-on expertise in developing ***full-stack applications*** and extracting actionable insights from ***complex data sets***. *Proven expertise in leveraging* ***programming and statistical tools*** *to solve complex problems and build scalable solutions*. Adept at working across all layers of the OSI model, offering a unique perspective on system and application-level challenges. *Strong analytical, communication, and problem-solving skills, with a focus on delivering high-quality,* ***data-driven insights and solutions****.*

TECHNICAL SKILLS

Java | Linux Shell | Docker | Python | Postgres | GitLab | OpenShift | Maven | Groovy | ReactJS | Javascript | CSS | AWS | GCP | Jenkins | Google BigQuery | Google Looker | Selenium | Katalon | Atlassian (JIRA, Confluence, Bitbucket) | C/C++

EDUCATION

Bellevue University – *Accelerated Bachelor's in Full Stack Web Development (In-Progress, started Oct. 2025)*
Cornell University – *Certificate in Data Science (2025) (certificate link)*
Colorado State University – *BS in Electrical Engineering with a Minor in Computer Science*

PROFESSIONAL HISTORY

Senior Software Engineer **June 2026 –August 2026**
ICF (Military Child Care)

- Fullstack development. Backend MS SQL Server Queries and Stored Procedures, Backend microservices, and helped fix UI issues associated with an upgrade from Angular 12 to Angular 20.

Senior Software Engineer **2025 – May 2026**
ICF (HHS CMS/Medicare BFD)

- Oversee the development, maintenance, and optimization of the Beneficiary Claims Data (BFD) system, a central component of Medicare/Medicaid. Maintain and enhance ancillary services for deployment, performance testing, and analytics (primarily in Python). Oversee AWS-hosted environments including Aurora PostgreSQL, EC2 instances, and AWS Lambda functions. Address technical debt, improve observability, and optimize performance and cost-efficiency. Terraform and OpenTofu. Cloud DevOps.

Principal Associate – Senior Software Engineer **2020 – 2025**
Guidehouse (FEMA National Flood Insurance Program - Pivot)

- Java, Spring, and Geospatial (DBeaver, PostGIS, and CoreLogic geocoding), used for flood risk rating Agile Software Used: Git, Atlassian (JIRA, Confluence), OpenShift, Jenkins, and Docker.
- Back-end development (Java and Spring REST API).
- Automated testing for geospatial service used for flood risk rating. Used Python and Java for test data generation in Bash env.
- Developed a Java application to hit REST API and produce hundreds of policies, quotes and claims from CSV data pulled from a production database and replace PII with dynamically generated random names and policy numbers for targeted testing (test data generation).
- Automated testing (Selenium and Katalon) for CI/CD environments and QA leadership.

Web Applications Developer II **2015-2018 & 2019-2020**
US Geological Survey

- Front-end and back-end development on a repository (http://www.sciencebase.gov) for scientific research and data for the USGS which is built from a variety of technologies and frameworks (Grails, Spring, Groovy, Java, Javascript, Bootstrap). Contributed to UI migration from Angular to ReactJS.
- Developed a full stack solution for metrics reporting on ScienceBase.gov site usage. Apache log parser written in Java deployed as a Docker image, PostgreSQL to store metrics, A Spring Boot REST API, and web-based reporting (charts, tables, and graphs) for usage metrics. Required designing and developing a relational database to store metrics and a scheduled service written in Java to parse the logs nightly and populate reporting tables. Web App UI dashboard for monitoring metrics incorporated Leaflet, Javascript, Bootstrap, Angular, ReactJS, and D3.

Web Applications Developer **2018 – 2019**
Automox

- AWS cloud-based OS (Linux, Windows, and MacOS) patching solution. Postgres, PHP Laravel backend code, REST API, frontend Javascript and bootstrap development, Golang agent improvements, and fixes to security update scripts.

Software Engineer R&D **2010 - 2015**
CA Technologies

- Java development for virtualization and storage monitoring probes and libraries. Development on next generation XenServer (Citrix XenServer REST API) and IBM PowerVM probes.
- Agile development practices and tools: Maven, Netbeans, Jenkins, Subversion (SVN), and Rally.
- Development of Nimsoft Monitor multi-platform installation and deployment using Java, python, and Ruby. Also required knowledge and support of MySQL, Oracle, and MS SQL Server databases.

Database Developer **2010**
Colorado State University DCE

- Performed SQL Server database upgrade/migration and developed custom SQL reports using T-SQL.

Systems Engineer VI **2000 - 2009**
Hewlett-Packard

- Developed web applications using MySQL, PHP, HTML, and JavaScript. Managed firmware code, system debugging, and system performance tuning for Intel-based workstations.

- EFI shell system administration and scripting
- R&D engineering representative for 64-bit (AMD64 and EM64T) graphics on Itanium-2 and x64 workstations
- Relationship management for 64-bit graphics drivers, firmware/EFI/BIOS, and hardware with ATI and NVIDIA
- Coordination and creation of 64-bit OS graphics test plans for Itanium-2 and x64 workstations (managed testers)
- DVI (Digital Video Interface) compliance qualification testing. Required expertise with electrical test harness and oscilloscope
- Technical Compute Farm (CF) cluster products (based on J6000 PA-RISC HP-UX workstations)
- Wrote UNIX scripts in Korn shell, Perl, C script, and various other scripting languages to manage CF clusters
- Licensing and integration for web-based (Big Brother) system management solution for HP CF clusters
- HP-UX system test and administration (Software Depot, SCM, UNIX scripting, C programming)